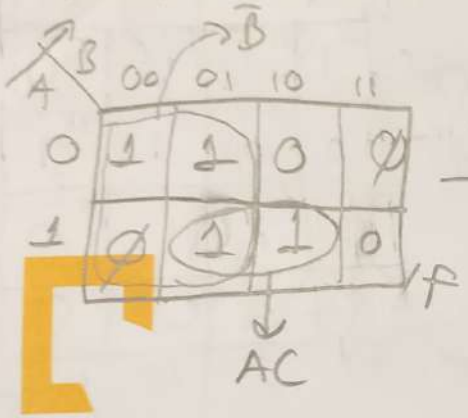
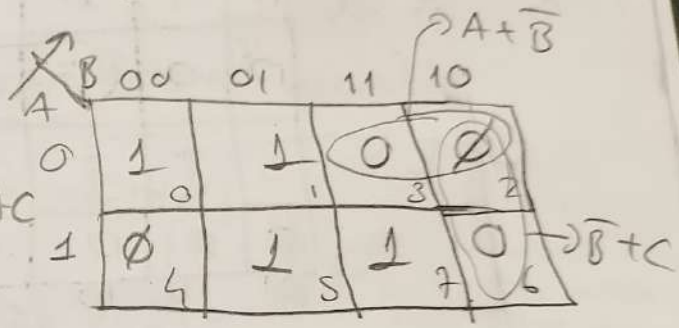
# Incompletly Specified Functions
## (DON'T CARES)

$f(A,B,C) = \Sigma(0,1,5,7) + \emptyset(2,4)$




$\rightarrow f_{SOP} = \bar{B} + AC$

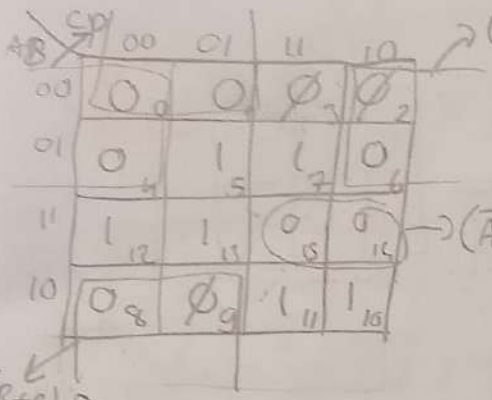


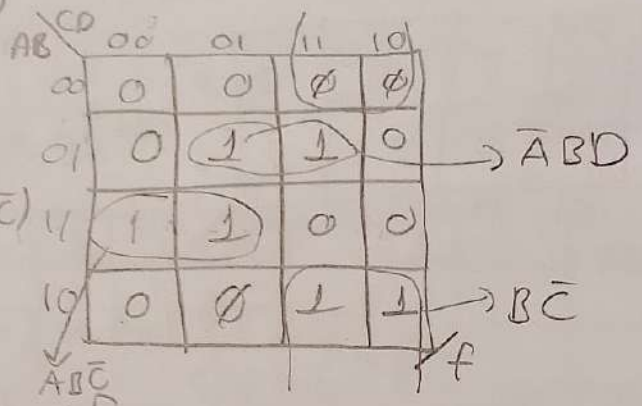


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$f_{POS} = (A + \bar{B}) \cdot (\bar{B} + C)$

$g(A,B,C,D) = \Pi M(0,1,4,6,8,12,15) \cdot \emptyset(2,3,9)$

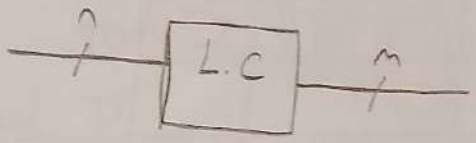



$f_{POS} = (B+C) \cdot (A+D) \cdot (\bar{A}+\bar{B}+\bar{C})$ $f_{SOP} = AB\bar{C} + \bar{B}C + \bar{A}BD$

---

Multiple Output Minimization



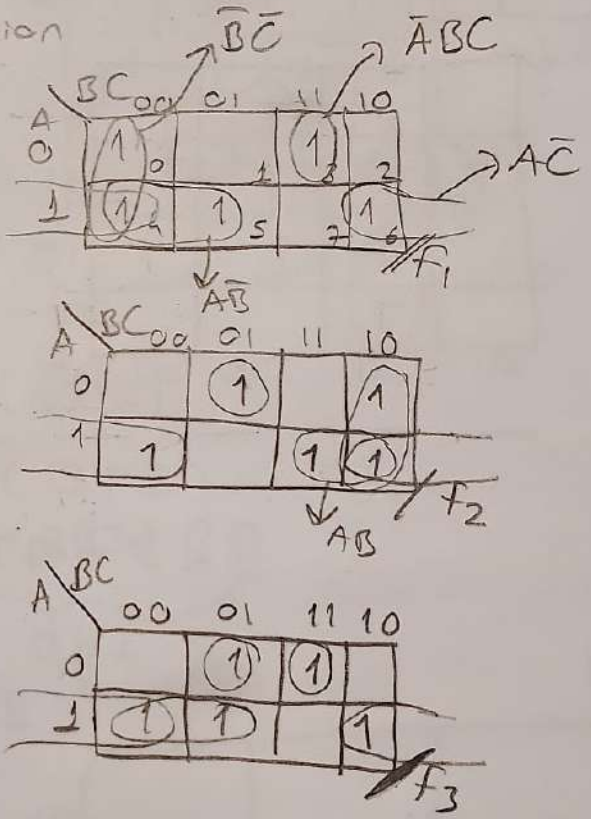

$f_1 = \Sigma_m(0,3,4,5,6)$

$f_2 = \Sigma_m(1,2,4,6,7)$

$f_3 = \Sigma_m(1,3,4,5,6)$

$f_1(A,B,C) = \Sigma$ ...

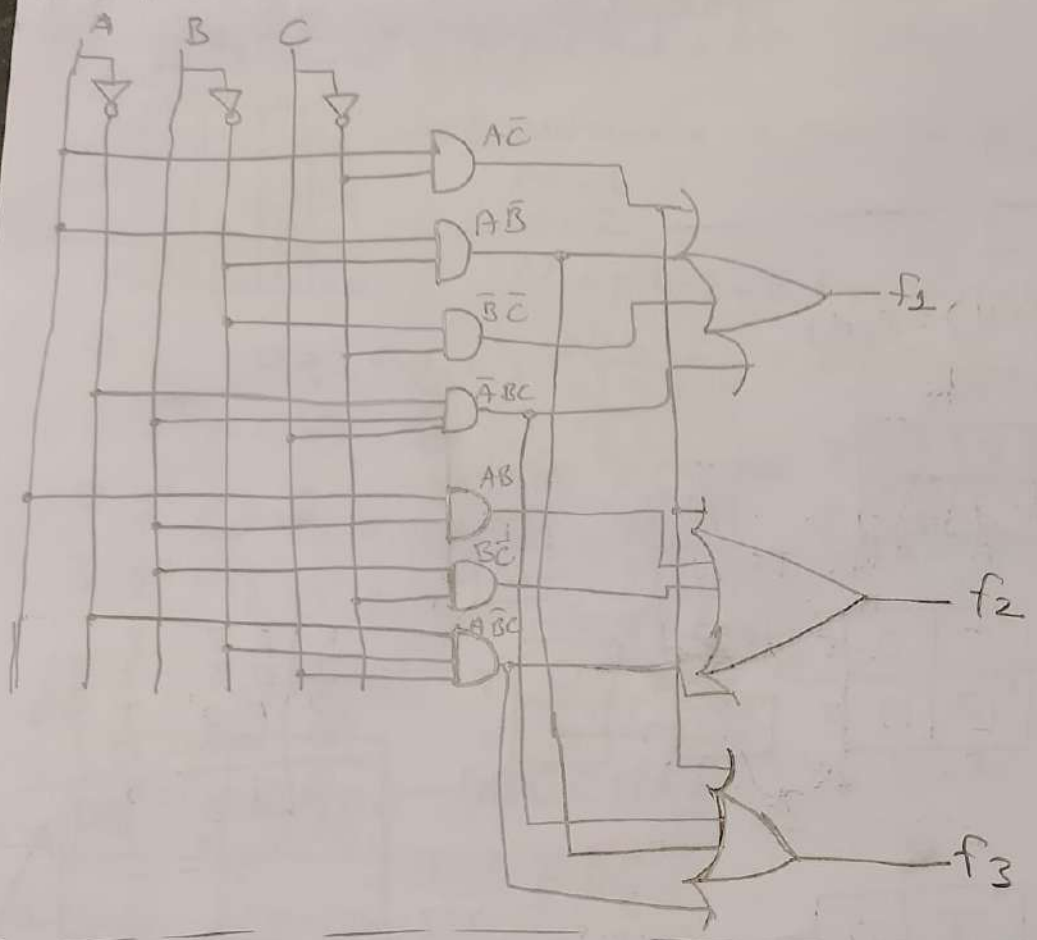
$f_1 = A\bar{C}$




$f_1 = A\bar{C} + A\bar{B} + \bar{B}\bar{C} + \bar{A}BC$

$f_2 = A\bar{C} + AB + B\bar{C} + \bar{A}\bar{B}C$

$f_3 = A\bar{C} + A\bar{B} + \bar{A}BC + \bar{A}\bar{B}C$




## ENTERED VARIABLE MAP REDUCTION

N: Input Variable

n: order of Map

$(n < N)$

$N - n$ : free variable = Entered variable

$2^{N-n}$ : Map Key, Truth Key

(Compressed Map Cell Number) · (Map Key) = Code Number of the first minterm or Maxterm possible for that Compressed map cell

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$f(A,B,C) = \Sigma_m(2,5,6,7)$

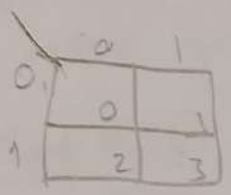
| Dec. | A | B | C | f | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | |
| 2 | 0 | 1 | 0 | 1 | $\bar{C}$ |
| 3 | 0 | 1 | 1 | 0 | |
| 4 | 1 | 0 | 0 | 0 | C |
| 5 | 1 | 0 | 1 | 1 | |
| 6 | 1 | 1 | 0 | 1 | |
| 7 | 1 | 1 | 1 | 1 | 1 |

$N = 3$

$n = 2$

$N - n = 3 - 2 = 1$ free variable : (C)

$2^{N-n} = 2^1 = 2$ (Map key)

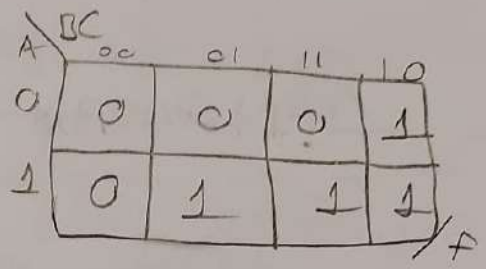



$0 \cdot 2 = 0 \rightarrow 0,1$

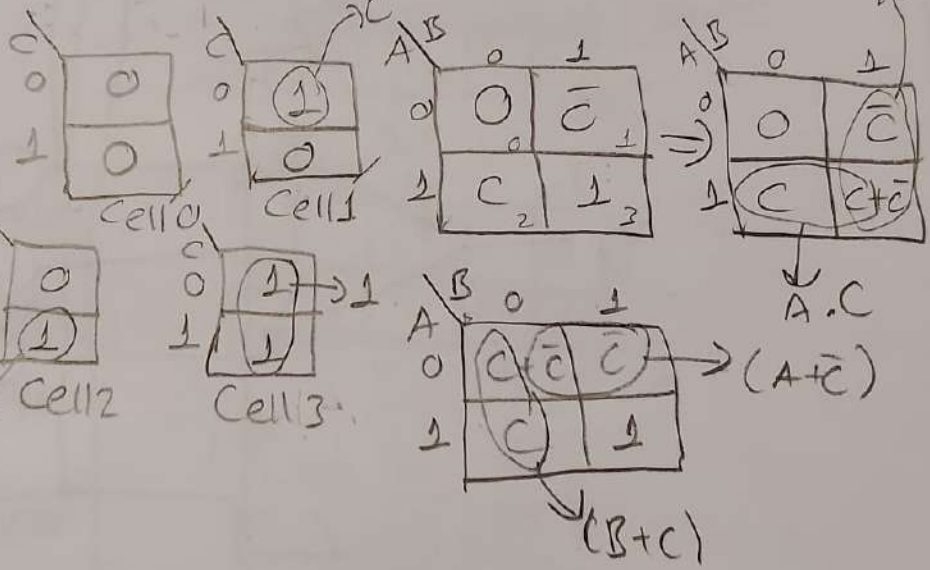
$1 \cdot 2 = 2 \rightarrow 2,3$

$2 \cdot 2 = 4 \rightarrow 4,5$

$3 \cdot 2 = 6 \rightarrow 6,7$

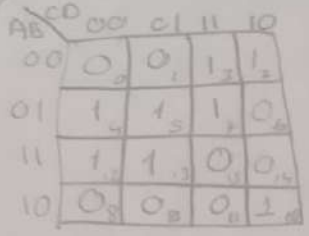



| Dec | A | B | f |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | $\bar{C}$ |
| 2 | 1 | 0 | C |
| 3 | 1 | 1 | 1 |




$fpos = (A+\bar{C}).(B+C)$

Örn: $f(A,B,C,D) = \Pi M(0,2,6,8,9,11,14,15)$



→ 2. Mertebede K-Map ile yapım

$N = 4$

$n = 2$

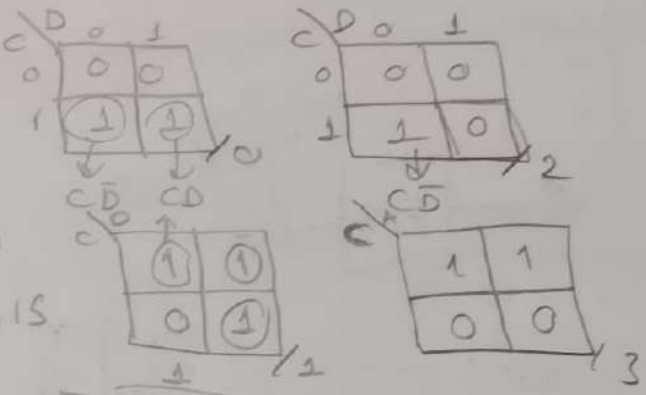
$N - n = 4 - 2 = 2$ free variables (C, D)

$0 \cdot 2^2 = 0 \Rightarrow 0,1,2,3$

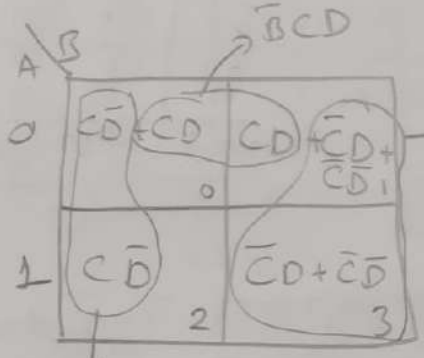
$1 \cdot 2^2 = 4 \Rightarrow 4,5,6,7$

$2 \cdot 2^2 = 8 \Rightarrow 8,9,10,11$

$3 \cdot 2^2 = 12 \Rightarrow 12,13,14,15$



$\bar{C}(D+\bar{D})$



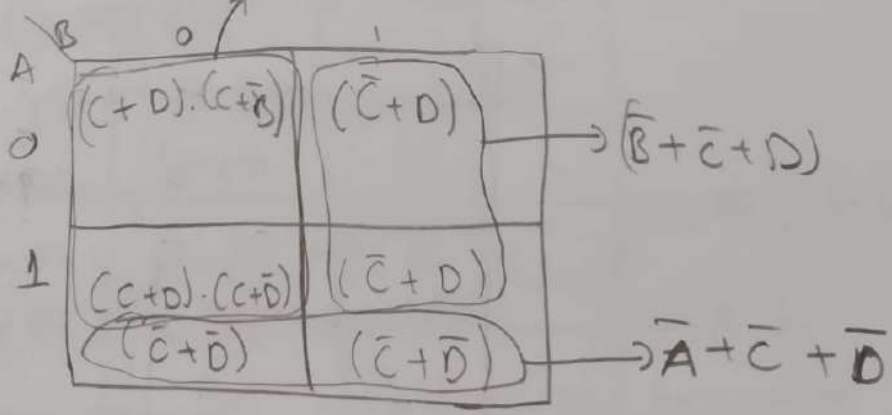
$\rightarrow B[\bar{C}D + \bar{C}\bar{D}]$

$= B\bar{C}$

$f_{sop} = B\bar{C} + \bar{B}C\bar{D} + \bar{A}CD$

$\bar{B}C\bar{D}$

$B + [(C+D).(C+\bar{D})]$



$\rightarrow (\bar{B}+\bar{C}+D)$

$\rightarrow \bar{A}+\bar{C}+\bar{D}$

$f_{pos} = (B+C).(\bar{B}+\bar{C}+D).(\bar{A}+\bar{C}+\bar{D})$

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Ör: 3 öğretmen 3 sorudan 40 40 20 puanlık bir sınav kağıdını 100 üzerinden değerlendiriyor. 1. ve 2. öğretmen 40 puanlık soruları 3. öğretmen ise 20 puanlık soruyu aşağıdaki şartlara göre değerlendiriyor.

1. öğretmen: 0, 15, 25, 40
2. öğretmen: 0, 20, 30, 40
3. öğretmen: 0, 20

Başarı Notu ≥ 50

Öğrencinin başarı durumunu gösteren devreyi NAND kapıları ve 4. mertebeden Karnaugh kapıları kullanarak gösteriniz.

1. öğretmen

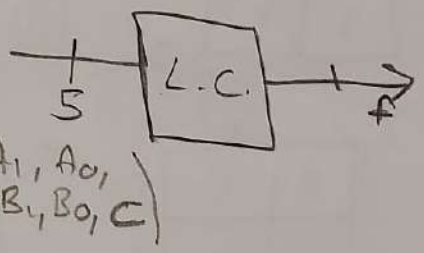
| $A_1$ | $A_0$ | puan |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 15 |
| 1 | 0 | 25 |
| 1 | 1 | 40 |

2. öğretmen

| $B_1$ | $B_0$ | puan |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 20 |
| 1 | 0 | 30 |
| 1 | 1 | 40 |

3. öğretmen

| C | puan |
|---|---|
| 0 | 0 |
| 1 | 20 |



| Der | | $A_1$ | $A_0$ | $B_1$ | $B_0$ | C | f |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 2 | 0 | 0 | 0 | 1 | 0 | 0 |
| | 3 | 0 | 0 | 0 | 1 | 1 | 0 |
| 2 | 4 | 0 | 0 | 1 | 0 | 0 | 0 |
| | 5 | 0 | 0 | 1 | 0 | 1 | 1 |
| 3 | 6 | 0 | 0 | 1 | 1 | 0 | 0 |
| | 7 | 0 | 0 | 1 | 1 | 1 | 1 |
| 4 | 8 | 0 | 1 | 0 | 0 | 0 | 0 |
| | 9 | 0 | 1 | 0 | 0 | 1 | 0 |
| 5 | 10 | 0 | 1 | 0 | 1 | 0 | 0 |
| | 11 | 0 | 1 | 0 | 1 | 1 | 1 |
| 6 | 12 | 0 | 1 | 1 | 0 | 0 | 0 |
| | 13 | 0 | 1 | 1 | 0 | 1 | 1 |
| 7 | 14 | 0 | 1 | 1 | 1 | 0 | 1 |
| | 15 | 0 | 1 | 1 | 1 | 1 | 1 |

| Der | | $A_1$ | $A_0$ | $B_1$ | $B_0$ | C | f |
|---|---|---|---|---|---|---|---|
| 8 | 16 | 1 | 0 | 0 | 0 | 0 | 0 |
| | 17 | 1 | 0 | 0 | 0 | 1 | 0 |
| 9 | 18 | 1 | 0 | 0 | 1 | 0 | 0 |
| | 19 | 1 | 0 | 0 | 1 | 1 | 1 |
| 10 | 20 | 1 | 0 | 1 | 0 | 0 | 1 |
| | 21 | 1 | 0 | 1 | 0 | 1 | 1 |
| 11 | 22 | 1 | 0 | 1 | 1 | 0 | 1 |
| | 23 | 1 | 0 | 1 | 1 | 1 | 1 |
| 12 | 24 | 1 | 1 | 0 | 0 | 0 | 0 |
| | 25 | 1 | 1 | 0 | 0 | 1 | 1 |
| 13 | 26 | 1 | 1 | 0 | 1 | 0 | 1 |
| | 27 | 1 | 1 | 0 | 1 | 1 | 1 |
| 14 | 28 | 1 | 1 | 1 | 0 | 0 | 1 |
| | 29 | 1 | 1 | 1 | 0 | 1 | 1 |
| 15 | 30 | 1 | 1 | 1 | 1 | 0 | 1 |
| | 31 | 1 | 1 | 1 | 1 | 1 | 1 |

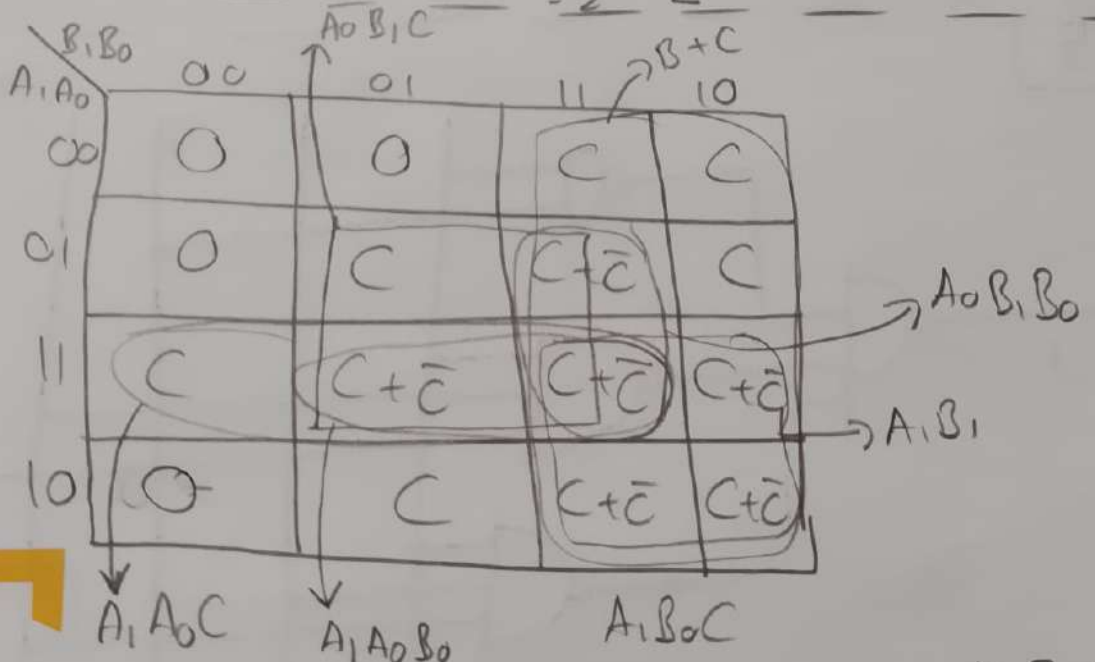
$N = 5$

$n = 4$

$N - n = 5 - 4 = 1$

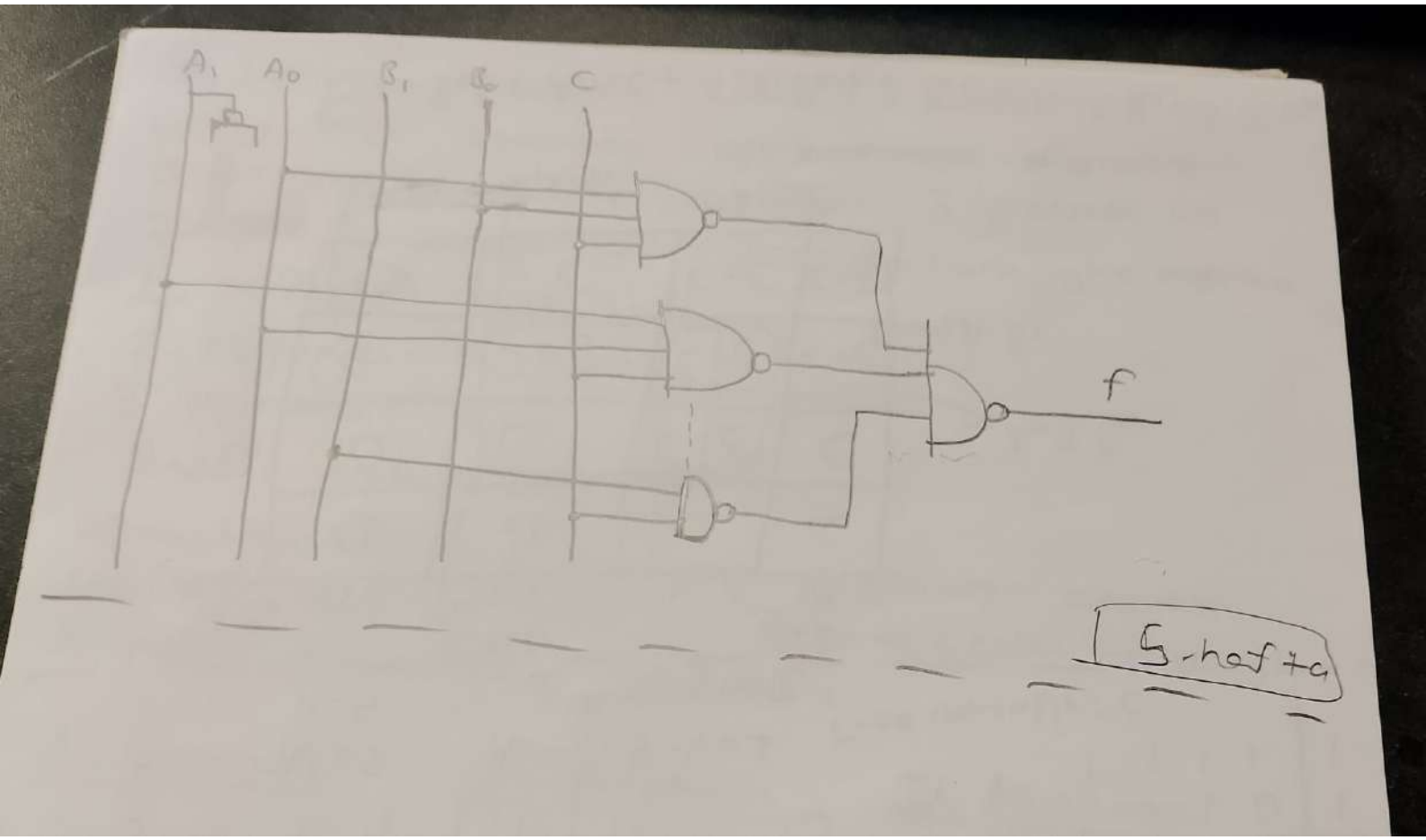
$2^{N-n} = 2^1 = 2$

Free variable : C




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$$f = A_0B_0C + A_1A_0C + A_0B_1B_0 + A_1B_0C + A_1A_0B_0 + A_1B_1 + B_1C$$




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